

MCKINNON ST SIMONS ISLAND, GA, USA

WMO: 722137

Lat: 31.152N

Lon: 81.391W

Elev: 5

StdP: 101.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	1.5	-9.5	1.7	3.7	-6.9	2.1	5.3	9.2	10.5	8.4	10.7	2.8	350	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	33.8	25.2	32.5	25.2	31.5	25.2	26.9	31.0	26.5	30.6	26.0	30.0	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	29.3	25.2	20.4	28.9	24.9	20.0	28.5	84.3	31.1	82.1	30.5	80.3	30.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.6	7.7	7.0		DB	-3.5	36.0	2.1	0.9	-5.0	36.6	-6.2	37.1	-7.4	37.6	-8.9	38.2
				WB	-5.4	27.9	1.9	0.6	-6.8	28.4	-7.9	28.8	-9.0	29.1	-10.4	29.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	11.1	12.9	15.9	19.7	23.9	26.9	28.0	27.7	25.9	21.5	16.3	13.1	(6)
	DBStd	6.93	4.89	4.59	4.27	3.35	2.62	1.69	1.37	1.54	2.11	3.61	4.26	4.70	(7)
	HDD10.0	104	47	21	7	0	0	0	0	0	0	0	4	24	(8)
	HDD18.3	778	227	158	95	24	1	0	0	0	0	14	88	171	(9)
	CDD10.0	3857	82	103	192	291	431	506	558	548	476	358	193	119	(10)
	CDD18.3	1488	2	6	21	66	174	256	300	290	226	113	26	8	(11)
	CDH23.3	12994	4	19	74	261	1163	2412	3233	3069	2008	666	77	8	(12)
	CDH26.7	4014	0	1	11	41	266	773	1211	1073	533	99	8	0	(13)
Wind	WSAvg	3.5	3.3	3.4	3.7	3.8	3.9	3.6	3.5	3.4	3.6	3.4	3.3	3.2	(14)
Precipitation	PrecAvg	1234	86	93	101	65	80	134	135	160	165	83	63	74	(15)
	PrecMax	1754	238	216	272	235	262	400	307	461	538	279	188	215	(16)
	PrecMin	867	9	18	12	5	7	19	26	30	2	0	6	0	(17)
	PrecStd	212	62	46	66	48	58	88	73	83	111	72	49	51	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	26.0	28.2	30.5	33.5	35.0	35.3	35.2	32.8	30.3	27.4	24.5	(19)
		MCWB	18.8	19.6	19.7	21.0	23.1	24.9	25.4	25.7	24.4	24.1	21.3	20.2	(20)
	2%	DB	21.2	23.0	25.3	27.5	31.0	33.1	33.7	33.2	31.2	28.9	25.3	22.6	(21)
		MCWB	17.2	18.2	18.5	20.6	22.5	24.7	25.5	25.9	24.6	23.4	20.6	19.3	(22)
	5%	DB	19.5	21.1	23.5	25.9	29.2	31.6	32.4	31.9	30.3	27.7	23.9	21.1	(23)
		MCWB	16.5	16.9	18.1	20.3	22.5	24.8	25.6	26.0	24.4	23.1	20.3	18.7	(24)
	10%	DB	18.0	19.2	21.9	24.7	28.0	30.3	31.3	30.9	29.4	26.7	22.6	19.6	(25)
		MCWB	15.7	16.2	17.5	19.8	22.4	24.6	25.5	25.6	24.2	22.5	19.6	17.8	(26)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	20.6	21.4	23.6	25.5	27.0	27.5	27.8	26.5	26.2	23.6	21.6	(27)
		MCDB	22.4	24.1	24.5	26.3	29.1	30.8	31.7	32.0	30.1	28.1	25.2	22.9	(28)
	2%	WB	18.4	19.2	20.2	22.6	24.6	26.3	26.9	26.9	25.8	24.9	22.4	20.3	(29)
		MCDB	20.2	21.8	23.3	25.3	28.0	30.3	31.2	31.0	29.2	27.5	23.9	21.6	(30)
	5%	WB	17.4	18.1	19.4	21.8	24.0	25.7	26.4	26.5	25.4	24.2	21.3	19.2	(31)
		MCDB	18.9	20.1	22.1	24.4	27.3	29.7	30.9	30.6	28.8	26.7	23.0	20.6	(32)
	10%	WB	16.2	17.0	18.5	20.9	23.4	25.2	25.9	26.0	25.0	23.5	20.1	18.0	(33)
		MCDB	17.8	18.7	21.1	23.6	26.6	29.1	30.1	29.9	28.2	25.9	22.1	19.4	(34)
Mean Daily Temperature Range		MDBR	9.2	9.0	8.9	8.3	7.7	7.3	7.3	6.7	6.4	7.9	9.0	8.7	(35)
	5% DB	MCDBR	9.7	10.1	10.2	9.1	9.5	9.1	8.6	7.9	7.2	7.4	9.0	8.9	(36)
		MCWBR	6.7	6.3	5.3	4.4	4.1	3.8	3.6	3.2	3.0	3.7	5.3	6.1	(37)
	5% WB	MCDBR	8.5	8.7	8.5	7.5	7.5	7.8	7.8	7.2	6.3	6.0	7.1	8.1	(38)
		MCWBR	6.6	6.2	5.2	4.5	3.6	3.8	3.6	3.3	3.0	3.4	5.1	6.2	(39)
Clear-Sky Solar Irradiance	taub	0.337	0.346	0.377	0.400	0.443	0.491	0.513	0.500	0.429	0.399	0.360	0.350	(40)	
	taud	2.530	2.508	2.438	2.365	2.280	2.200	2.147	2.189	2.386	2.433	2.496	2.518	(41)	
	Ebn at Noon	892	916	906	893	852	807	787	794	843	846	857	856	(42)	
	Edh at Noon	86	96	111	123	135	146	153	145	115	103	89	84	(43)	
All-Sky Solar Radiation	RadAvg	3.04	3.65	4.83	5.98	6.46	6.19	6.19	5.63	4.76	4.19	3.35	2.70	(44)	
	RadStd	0.19	0.40	0.45	0.38	0.37	0.46	0.37	0.36	0.40	0.40	0.20	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.72	+1.43	N/A	+0.40	N/A	N/A	-33	-132	+247	+163
(47)	Regional (29 neighbors)	+0.49	+0.76	N/A	N/A	N/A	N/A	N/A	-73	+167	+120

Nomenclature: See separate page